cognizancejournal.com
Marie Sofia A. Gotinga *et al*, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.5, May 2025, pg. 598-606
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

Insulin Plant (*Costus igneus*) Tea as an Alternative Hypoglycemic Medication

Marie Sofia A. Gotinga[1]; Khinzly Harry M. Mascardo[2]; Lyuris Rosal E. Malaza[3]; Rica Eve Y. Tamara[4]; Herlaw G. Lito[5]; Kent Lloyd M. Destajo[6]

[123456]Basic Education Department, St. Paul University Surigao, Philippines
[1] mariesofia.gotinga@spus.edu.ph; [2] khinzlymascardo9@gmail.com; [3] lyurisrosalmalaza@gmail.com; [4] rrcaeytmraa@gmail.com; [5] herlawlito1@gmail.com; [6] kentlloyd.destajo@spus.edu.ph

[1]ORCID: 0009-0006-5780-4431
[2]ORCID: 0009-0007-4606-0251
[3]ORCID: 0009-0009-7780-7447
[4]ORCID: 0009-0003-7931-1740
[5]ORCID: 0009-0004-6926-5944
[6]ORCID: 0009-0009-7616-6203

DOI: 10.47760/cognizance.2025.v05i05.038

Abstract: The research investigates the potential of Costus igneus, commonly known as the insulin plant, as an alternative hypoglycemic medication. The study employs a developmental research design to evaluate the acceptability and effectiveness of Costus igneus tea in managing hyperglycemia. The methodology includes five phases: gathering of Costus igneus leaves, preparation of Costus igneus leaves, packaging of dried Costus igneus leaves, production of Costus igneus tea, and sensory evaluation. To assess the acceptability of Costus igneus tea as a convenient alternative for commercial hypoglycemic medication, the researchers specifically conducted a 9-point Hedonic Rating Scale and Descriptive Rating Test with fifteen (15) evaluators consisting of TLE teachers, science teachers, and parents. The findings indicate that the visual presentation and sensory attributes of the tea significantly influence consumer acceptability, suggesting that the appearance of the product plays a crucial role in its marketability. The results demonstrate that Costus igneus tea is not only a viable alternative for individuals with hyperglycemia but also possesses favorable sensory qualities that enhance its appeal. The study highlights the importance of traditional herbal remedies in contemporary health practices and encourages further exploration of Costus igneus as a natural hypoglycemic agent. Overall, this research contributes to the growing body of evidence supporting the use of herbal teas in diabetes management, emphasizing the need for more accessible and natural treatment options.

Keywords: Alternative medicine, Costus igneus, herbal tea, hypoglycemic medication, sensory evaluation

I. INTRODUCTION

Diabetes has continuously increased over time and has affected many people across the world. According to the IDF Diabetes Atlas (2021), among adults aged 20 to 79 years of age, 10.5% have diabetes. Between the years 2000 and 2019, there has been a 3% increase in age-specific diabetic-related deaths (World Health Organization: Diabetes, 2023). Due to the cost of treatment, people end up not receiving the proper treatment for their condition (Jimeno et al., 2021). Therefore, it is essential to determine a safe, efficient, and cost-effective alternative to conventional medication.

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

cognizancejournal.com
Marie Sofia A. Gotinga *et al*, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.5, May 2025, pg. 598-606
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

In this context, Rahman et al. (2022) stated that plant-based medications are believed to be at the top of the list as a preventive measure for diseases and their related complications. Plant-based medication, also known as herbal medicine, is not a new treatment since, from generation to generation, the knowledge about herbal medicine has been practiced worldwide. Over time, various herbal medicines have been used, developed, and discovered despite their risks (Tolentino et al., 2019). As stated by Pang et al. (2019), herbal medicine is a good complementary and alternative treatment for type 2 diabetes mellitus (T2DM) because of its holistic quality that is capable of treating T2DM in various ways such as through anti-inflammatory, anti-oxidation, blood lipid regulation, and anti-glucose properties. Additionally, tea contains bioactive properties such as flavonoids and alkaloids, which have effects that lower blood glucose levels.

According to Zhang et al. (2022), multiple studies have shown that herbal tea has effectively improved glucose metabolism disorders in individuals with type 2 diabetes mellitus (T2 DM). One traditional plant that has recently garnered global acceptance is the insulin plant. The herbaceous plant named *Costus igneus* (insulin plant), belonging to the family Costaceae and native to Central and South America, is commonly referred to as the "insulin plant" because of its properties that aid in building up the insulin in the human body by strengthening β-cells of the pancreas in the body (Godi, 2019 and Vijay et al., 2023). Diabetic people have consumed the leaves of this plant because it is believed to lower blood glucose levels, which has been confirmed by diabetic people in the past to truly decrease an individual's blood glucose levels (Vijay et al., 2023).

Thus, in this study, the researchers will create a *Costus igneus* tea and test its overall characteristics: appearance, aroma, taste, and texture, if they are recommendable as a convenient way of consumption. Moreover, it is recommended as an alternative hypoglycemic medication. However, it is merely intended to be an alternative form of ready-made hypoglycemic medication and not a replacement for it. The results of this study not only focus on those individuals who suffer from hyperglycemia but also on everyone, especially those who wish to avoid this condition or any other complications caused by hyperglycemia.

II. METHODOLOGY

The study used a development research design through a quantitative approach. The developmental research design facilitated the systematic creation and refinement of the herbal tea. The quantitative research allowed for the objective measurement and statistical analysis of the tea's characteristics using numerical data. The study was conducted in four (4) phases: (1) Preparation of *Costus igneus* (insulin plant) leaves, (2) *Costus igneus* tea production, (3) Basic Nutritional Facts Analysis, and (4) Sensory evaluation. The study aimed to provide the characteristics of the resulting product in terms of appearance, aroma, taste, and texture of *Costus igneus* tea as an alternative hypoglycemic medication.

In this study, the researchers used an adapted questionnaire from Romano et al. (2024) divided into two (2) parts. Part I was the Hedonic Rating Scale, in which a numerical rating from 1 to 9 will determine the rate of acceptability of the product, in which 1 indicates "*Dislike Extremely*" and 9 indicates "*Like Extremely*". Additionally, the researchers also used the Descriptive Rating Test (Part II) with a 4-point scale with 1 indicating "*Highly Not Recommendable*" and 4 as "*Highly Recommendable*"; this was used to provide a detailed description of the product's sensory evaluation.

This study used quota and purposive sampling to select evaluators, including food experts, TLE teachers, and science teachers of St. Paul University Surigao. This sampling method ensured an accurate group of evaluators was selected to participate in this study in order to gather relevant responses about the product. Before participating, they were provided with relevant information about the evaluation process and gave their consent to assess the product.

III. RESULTS AND DISCUSSION

This chapter presents the results and discusses the data obtained from the responses of fifteen (15) respondents. The data were carefully analyzed, presented, discussed, and interpreted according to the perimeter of this research to answer the problem sought for this study.

Parameters	Units	Value	Methods
Moisture	%	4.70	Vacuum Oven Drying
Protein	%	12.0	Kjeldahl

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

cognizancejournal.com
Marie Sofia A. Gotinga *et al*, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.5, May 2025, pg. 598-606
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

Fiber	%	24.5	Ankom
Fat	%	2.78	Mojonnier Extraction
Ash	%	10.2	Ignition – Gravimetric
Carbohydrates	%	70.4	By Computation
Total Sugar	%	.0848	Munson-Walker Method

Table 1. *Basic Nutritional Facts Analysis*

The nutritional facts analysis of *Costus igneus* (insulin plant) tea are listed in Table 1. The parameters are described from largest to smallest, along with certain essential information about the product's nutritional value and how it will affect the health of the consumer and market demands. Based on the results, it can be observed that *Costus igneus* tea has a high carbohydrate content of 70.4%, making it a potential energy source for those relying on carbohydrates for daily fuel. Despite this, it contains bioactive compounds like corosolic acid and flavonoids that enhance insulin sensitivity, implying hypoglycemic benefits. This suggests that the tea could support energy needs while aiding in blood sugar regulation, which is particularly relevant in the Philippines, where carbohydrate-rich diets are standard. Clemente-Suárez et al. (2022) state that foods high in carbohydrates are an important part of a healthy diet since they provide the body with glucose to support bodily functions and physical activity. This finding was also supported by Mohanta et al. (2022), who stated that herbal materials offer a rich source of structurally diverse and bioactive polysaccharides.

The protein content of *Costus igneus* (insulin plant) tea is considerably high at 12.0%, determined using the Kjeldahl method. This suggests that the tea is a good plant-based protein source, supporting muscle maintenance and overall health. Its high protein value implies potential benefits for individuals shifting to a plant-based diet, offering an alternative to animal protein sources. In their article, Ma et al. (2022) stated that consumers are moving towards a more plant-based diet. However, some consumers avoid common plant proteins such as soy and gluten due to their potential allergenicity. Therefore, alternative protein sources are being explored as functional ingredients in foods. This implies that this product could be a novel potential source of protein. The fiber content with a value of 24.5% was determined by an Ankom method. This presents the dietary benefits accruable to the formulation. As fiber is essential in the human gut, this product impacts the body's glucose level, improves gut health, and prevents chronic disease. In line with these findings, Xiet et al. (2020) explained that soluble dietary fiber is prompted as an important part of reducing blood glucose, ameliorating insulin resistance and controlling body weight.

Additionally, the ash content of *Costus igneus* (insulin plant) tea is 10.2%, indicating a rich mineral profile. This reflects the presence of essential minerals that contribute to overall health and well-being. Its high mineral content implies potential benefits for maintaining electrolyte balance and supporting various bodily functions. As noted in the findings of Mayer et al. (202), micronutrient malnutrition is widespread and is linked with diets low in fruit and vegetables. This signifies that a need for a convenient and healthy drinkable product that can be available everywhere may aid this common problem. In this context, Ceccanti et al. (2021) added that minerals in the human diet are essential for metabolism and human health.

The moisture content of *Costus igneus* tea is 4.70%, determined using the vacuum oven method. This method involves applying high temperatures under reduced pressure to remove moisture effectively. The low moisture content implies better shelf stability and reduced risk of microbial growth, enhancing the product's storage life. Considering this, Zhang et al. (2022) justified that heat treatment is one of the most important methods to improve foodstuffs' physicochemical properties and nutritional profile, resulting from changes in protein structure and protein denaturation. Furthermore, the total sugar content of *Costus igneus* (insulin plant) tea is the lowest among its components at 0.0848%, measured using the Munson-Walker Method. This low sugar level suggests that the tea is suitable for individuals monitoring their sugar intake, such as those with diabetes. It also implies that the tea can provide energy without significantly impacting blood glucose levels.

Low sugar is desirable because a high intake of sugars is associated with numerous diseases (WHO, 2022); the organization also reported that excessive sugar intake, particularly in the form of sugar-sweetened beverages (SSBs), contributes to various health risks, including obesity, dental caries, and chronic diseases. This result makes the product appealing to diabetic and calorie-conscious individuals pursuing low-sugar snacks.

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

cognizancejournal.com
Marie Sofia A. Gotinga *et al*, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.5, May 2025, pg. 598-606
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

Variable	**M**	**SD**	**I**
Appearance	8.13	0.83	LE
Aroma	7.67	0.90	LVM
Taste	7.47	0.83	LVM
Texture	7.60	1.35	LVM
Average	7.72	0.79	LVM

Table 2. *General Acceptability of Costus igneus Tea through Hedonic Rating Scale*

Scale	***Interval***	***Interpretation***
9	*8.12 - 9.00*	*Like Extremely (LE)*
8	*7.23 - 8.11*	*Like Very Much (LVM)*
7	*6.34 - 7.22*	*Like Moderately (LM)*
6	*5.45 - 6.33*	*Like Slightly (LS)*
5	*4.56 - 5.44*	*Neither Like nor Dislike (NLD)*
4	*3.67 - 4.55*	*Dislike Slightly (DS)*
3	*2.78 - 3.66*	*Dislike Moderately (DM)*
2	*1.89 - 2.77*	*Dislike Very Much (DVM)*
1	*1.00 - 1.88*	*Dislike Extremely (DE)*

Table 2 presents the general acceptability of *Costus igneus* (insulin plant) tea through a hedonic rating scale. Based on the findings, Appearance got the highest score with a mean of 8.13 and a standard deviation of 0.83, which can be interpreted as "*Like* Table 2 presents the general acceptability of *Costus igneus* (insulin plant) tea through a hedonic rating scale. Based on the findings, Appearance got the highest score with a mean of 8.13 and a standard deviation of 0.83, which can be interpreted as "*Like Extremely*" (LE). This means that since the visual presentation of the tea was highly preferred by consumers, it indicated that the product's appearance contributed greatly to its acceptability. These findings support the results of Wang et al. (2024) that the findings established a quantitative evaluation model for the appearance of needle-shaped green teas based on image features. In addition, Song et al. (2021) further stated that fixing is the first and key step for forming green tea's specific appearance and fragrance. With traditional methods, only trained experts with years of experience can produce good green tea products, which is backward in the industrial age

Aroma's average score was 7.67 (SD = 0.90), which corresponds to "*Like Very Much*" (LVM). This suggests that while the aroma was generally well-received, the pemphis scent may have affected its acceptance compared to other sensory attributes, as emphasized by the evaluators. Given its unique aroma and refreshing taste, green tea is keenly appreciated by consumers (Xie et al., 2023). The researchers also noted that aroma is important in evaluating tea quality. It has a vital influence on consumers' choices.

Taste received a mean score of 7.47 and a standard deviation of 0.83, interpreted as "*Like Very Much*" (LVM), indicating a positive flavor acceptance. The standard deviation of 0.83 suggests moderate variability in respondents' ratings, meaning that while most consumers liked the taste, there were differences in individual preferences. This indicates potential for improvement to achieve a more consistent and higher level of acceptability.

Evaluators also highlighted that there is a slight bitterness in the tea. This finding is supported by Ye et al. (2022) that bitterness and astringency of tea are evoked by secondary metabolites present in tea leaves. The buds and tender leaves of tea plants are superior raw materials for producing beverages with unique flavors and health benefits. Feng et al. (2024) further explained that tea's popularity and flavor are influenced by factors like cultivation, processing methods, and shaping techniques, which also impact tea flavor.

Furthermore, texture was rated a mean of 7.60 (SD = 1.35), interpreted as "*Like Very Much*" (LVM), which demonstrated that the mouthfeel was pleasant, but refinement was needed that could be improved. As per the evaluator, the tea product is dry, showing processed dry powder. However, drying is part of the process. This was also highlighted by Sheng et al. (2023), who stated that drying is an important process in black tea processing; meanwhile, the moisture content is an important factor in determining the drying quality. However, there is a lack of effective real-time detection methods for the moisture content of black tea during drying.

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

cognizancejournal.com
Marie Sofia A. Gotinga *et al*, Cognizance Journal of Multidisciplinary Studies, Vol.5, Issue.5, May 2025, pg. 598-606
(An Open Accessible, Multidisciplinary, Fully Refereed and Peer Reviewed Journal)
ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

Variable	**M**	**SD**	**I**
Appearance	3.73	0.46	SA
Aroma	3.53	0.52	SA
Taste	3.20	0.68	A
Texture	3.27	0.80	SA
Average	3.43	0.47	SA

Table 3. *General Acceptability of Costus igneus Tea through Hedonic Rating Scale*

Scale	***Parameters***	***Verbal Interpretation***
4	*3.25-4.00*	*Strongly Agree (SA)*
3	*2.50-3.24*	*Agree (A)*
2	*1.75-2.49*	*Disagree (D)*
1	*1.00-1.74*	*Strongly Disagree (SD)*

Table 3 shows the general acceptability of insulin plant tea using a descriptive rating test. Appearance received the highest rating with a mean score of 3.73 and a standard deviation of 0.46, classified as "*Strongly Acceptable*" (SA). This suggests that the tea's visual appeal significantly contributed to its positive perception, indicating that the product is visually well-received and likely recommendable based on appearance alone. Song et al. (2021) noted that appearance and flavor are crucial indicators of the tea's quality. Hence, having an appearance as the most favored among all variables indicates that the product has a good advantage if pursued to be sold in the market. However, Li et al. (2022) suggested that the dark color of some green teas can be improved through technological experiments in the future.

The next highest mean score was for aroma (3.53, SD = 0.52), also considered "*Strongly Acceptable*," indicating that the tea was favorably received in terms of odor, albeit with less visual impact than previous results. Given the positive reception of its aroma, the product is recommended as it appeals to consumers' sensory preferences. The findings are consistent with Romano et al. (2024) that aroma, scoring 3.50, is designated as *Highly Recommended* (HR) and contributes a mild yet pleasant scent, aligning with the subtle nature of herbs. As Song et al. (2024) conducted, different scented teas provide various choices for consumers in appearance, aroma, flavor, and others.

The mean for taste was 3.20 with a standard deviation of 0.68, rated as "*Acceptable*" (A), indicating that the flavor of the tea met basic expectations but could be improved to enhance its appeal. Although there is room for enhancement, the product is still recommendable based on its overall acceptability. Based on the study conducted by Wang, Yue, & Tong (2021) showed that 22 chemical components had an important contribution to the taste characteristics of yellow tea, and the chemical components that had an important influence on each taste component were obtained.

Texture received a mean score of 3.27 with a standard deviation of 0.80, indicating that the tea's mouthfeel was generally well-received and classified as "*Strongly Acceptable*." The relatively high acceptance suggests that the texture contributed positively to the overall sensory experience. Based on this result, the product is recommendable in terms of texture. The findings agree with that of Ambid et al. (2024) that texture is the least favored characteristic of the developed product. However, it still falls within the recommended category. The global average acceptability was 3.43 (SD = 0.47), classifying the tea as "*Strongly Acceptable*," revealing the characteristics of tea in terms of smell and taste consumed, but with room for improvement of the sensory profile of the product in order to enhance consumer acceptance.

IV. CONCLUSIONS

In conclusion, *Costus igneus* (insulin plant) is accepted and recommended as an alternative tea for hypoglycaemic medication. Among all attributes, appearance was the most accepted and recommended, texture was the least accepted, and taste was the least recommended.

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

ISSN: 0976-7797
Impact Factor: 5.183
Index Copernicus Value (ICV) = 77.57

ACKNOWLEDGEMENT

The researchers are profoundly thankful to their parents for their unwavering love, endless support, and constant encouragement. The belief their parents have shown in them has served as a guiding light, helping them find strength through every challenge and joy in every success. The researchers acknowledge that they are who they are today because of the love and sacrifices of their parents.

REFERENCES

1. Adebayo, P. O., Omobuwa, B. K., & Adebayo, H. A. (2021). Hypoglycemic and antioxidant activities of costus speciosus leaf extract in alloxan-induced diabetic rats. *Journal of Ethnopharmacology, 263,* 113222. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3931203/
2. Alqathama, A., Alluhiabi, G., Baghdadi, H., Aljahani, L., Khan, O., Jabal, S., Makkawi, S., & Alhomoud, F. (2020). Herbal medicine from the perspective of type II diabetic patients and physicians: What is the relationship? *BMC Complementary Medicine and Therapies, 20*(1). https://doi.org/10.1186/s12906-020-2854-4
3. The Old Farmers Almanac. (n.d.). *Drying your own herbs for tea.* Retrieved December 3, 2024, from https://www.almanac.com/drying-your-own-herbs-tea
4. Ambid, C. J. P., Ocampo, J. L. A., Requillo, D. B., Butao, K. J. D., Destajo, K. L. M., & Dumadag, E. J. D. L. (2024). Portunus pelagicus (blue manna crab) shells as polvoron. *Cognizance Journal of Multidisciplinary Studies, 4*(5), 134–141. https://doi.org/10.47760/cognizance.2024.v04i05.012
5. American Diabetes Association. (2022, January 15). *Hypoglycemia (low blood sugar).* https://www.diabetes.org/diabetes/medication-management/blood-glucose-testing-and-control/hypoglycemia
6. Arivu, I., Muthulingam, M., & Selvakumar, G. (2023). Detailed study on Costus igneus plant for its medicinal importance - A review. *International Journal of Zoology and Applied Biosciences, 8*(1), 34–39. https://doi.org/10.55126/ijzab.2023.v08.i01.005
7. Athilli, L., Siddiqui, A. F., Hussain, F., & Parvez, E. A. H. (2021). Pharmacognostic study and pharmacological potentials of costus igneus plant–review. *International Journal of Pharmacognaosy, 8*(12), 476-486. https://doi.org/10.13040/ijpsr.0975-8232.ijp.8(12).476-86
8. Brimson, J. M., Prasanth, M. I., Malar, D. S., Sharika, R., Sivamaruthi, B. S., Kesika, P., Chaiyasut, C., Tencomnao, T., & Prasansuklab, A. (2021). Role of herbal teas in regulating cellular homeostasis and autophagy and their implications in regulating overall health. *Nutrients, 13*(7), 2162. https://doi.org/10.3390/nu13072162
9. Brutsaert, E. F. (2024, March 18). *Diabetes mellitus (DM).* MSD Manual Consumer Version. https://www.msdmanuals.com/home/hormonal-and-metabolic-disorders/diabetes-mellitus-dm-and-disorders-of-blood-sugar-metabolism/diabetes-mellitus-dm
10. Chandrasekara, A., Herath, A., Abeynayake, R., De Camargo, A. C., & Shahidi, F. (2021). Synergies of herbal teas on improved plasma antioxidant capacity and postprandial hypoglycemic response in healthy adults. *Journal of Food Bioactives, 13.* https://doi.org/10.31665/jfb.2020.13262
11. Capillas, R. C. & Herrero, A. N. (2021 March 3) Sensory analysis and consumer research in new product development. *Foods, 10* (3), 582. https://doi.org/10.3390/foods10030582.
12. Chen, H., & Lu, Y. (2021). An overview of hypoglycemic modern drugs. In Chen, H., & Zhang, M. (Eds.), *Structure and Health Effects of Natural Products on Diabetes Mellitus.* (pp. 17-32). Springer https://doi.org/10.1007/978-981-15-8791-7_2
13. Ceccanti, C., Brizzi, A., Landi, M., Incrocci, L., Pardossi, A., & Guidi, L. (2020). Evaluation of major minerals and trace elements in wild and domesticated edible herbs traditionally used in the Mediterranean area. *Biological Trace Element Research.* https://doi.org/10.1007/s12011-020-02467-3
14. Clemente-Suárez, V. J., Mielgo-Ayuso, J., Martín-Rodríguez, A., Ramos-Campo, D. J., Redondo-Flórez, L., & Tornero-Aguilera, J. F. (2022). The burden of carbohydrates in health and disease. *Nutrients, 14*(18), 3809. https://doi.org/10.3390/nu14183809
15. Davis, H. A., Spanakis, E. K., Cryer, P. E., & Davis, S. N. (2021, June 29). *Hypoglycemia during therapy of diabetes.* Endotext. https://www.ncbi.nlm.nih.gov/books/NBK279100/
16. Ekanem, I., Antia, B., Essien, E., & Thomas, P. (2023). In vitro antioxidant activity and hypoglycemic efficacy of the leaf, stem, and rhizome extracts of costus igneus nak (Costaceae) in alloxan induced diabetic rats. *World Journal of Applied Science & Technology, 15*(1), 98–103. https://doi.org/10.4314/wojast.v15i1.98
17. Feng, J., Tian, W., Wang, J., Ye, S., Pan, G., Yu, B., & Hao, Z. (2024). The shaping of milk-flavored white tea: more than a change in appearance. *Journal of Integrative Agriculture, 23*(11), 3912-3922. https://doi.org/10.1016/j.jia.2024.09.010
18. Harler, & Ronald, C. (2025, March 13). *Tea plant. Encyclopedia Britannica*. https://www.britannica.com/plant/tea-plant

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

19. Heller, S., Peyrot, M., Oates, S. K., & Taylor, A. D. (2020). Hypoglycemia in patient with type 2 diabetes treated with insulin: it can happen. *BMJ Open Diabetes Research and Care, 8*(1), e001194. https://doi.org/10.1136/bmjdrc-2020-001194
20. *How insulin and glucagon regulate blood sugar*. (2023, February 15). Medical News Today. https://www.medicalnewstoday.com/articles/316427#summary
21. *How to grind tea leaves (the 6 methods).* (n.d.). BaristaJoy. Retrieved February 12, 2024, from https://baristajoy.com/grind-tea-leaves/
22. *How to steep loose leaf tea to get the perfect cup of tea.* (2022, January 23). Justea. https://justea.com/blogs/tea-101/how-to-brew-tea
23. International Diabetes Federation. (2024, February 7). *Diabetes facets and figures.* https://idf.org/about-diabetes/diabetes-facts-figures/
24. Jimeno, C., Sy, R. A., De La Pena, P., Cipriano, C., Tan, R., Panelo, A., & Ng, J. (2021). Direct medical costs of type 2 diabetes mellitus in the Philippines: findings from two hospital databases and physician surveys. *BMJ Open, 11*(10), e049737. https://doi.org/10.1136/bmjopen-2021-049737
25. JusTea, justly made tea. (n.d.) Justea. Retrieved Dcember 15, 2024, from https://justea.com/
26. Kai, K., MA, Greis, M., Lu, J., Nolden, A. A., McClements, D. J., & Kinchla, A. J. (2022). Functional performance of plant proteins. *Foods, 11*(4), 594. https://doi.org/10.3390/foods11040594
27. Kumar, S., Mittal, A., Babu, D., & Mittal, A. (2020). Herbal medicines for diabetes management and its secondary complications. *Current Diabetes Reviews, 17*(4), 437–456. https://doi.org/10.2174/1573399816666201103143225
28. Li, Y., Ran, W., He, C., Zhou, J., Chen, Y., Yu, Z., & Ni, D. (2022). Effects of different tea tree varieties on the color, aroma, and taste of Chinese Enshi green tea. *Food Chemistry X, 14*, 100289. https://doi.org/10.1016/j.fochx.2022.100289
29. Mau, S., Saleem, S., Kimothi, V., Joshi, V. V., & Singh, S. (2019). An exhaustive review on herbs used in the management of diabetes mellititus type 2. *Journal of Applied Pharmaceutical Sciences and Research,* 5–11. https://doi.org/10.31069/japsr.v2i2.2
30. McGrowder, D. A., Miller, F. G., Nwokocha, C., Wilson-Clarke, C. F., Anderson, M., Anderson-Jackson, L., Williams, L., & Alexander-Lindo, R. (2021). Medical herbs and the treatment of diabetes mellitus: Mechanisms of action. In A. Hussain & S. Behl (Eds.), *Treating endocrine and metabolic disorders with herbal medicines* (pp. 48-73). IGI Global Scientific Publishing. https://doi.org/10.4018/978-1-7998-4808-0.ch003
31. Ma, K. K., Greis, M., Lu, J., Nolden, A. A., McClements, D. J., & Kinchla, A. J. (2022). Functional performance of plant proteins. *Foods (Basel, Switzerland), 11*(4), 594. https://doi.org/10.3390/foods11040594
32. Mathew, F., & Varghese, B. (2019). On medicinal exploration of Costus igneus: The insulin plant C review. *International Journal of Pharmaceutical Sciences Review and Research.* https://www.semanticscholar.org/paper/on-Medicinal-Exploration-of-Costus-igneus-%3A-The-C-Mathew-Varghese/a80280d2dc12c4c1b6d48737d0a8980bad136117
33. Mayer, A. B., Trenchard, L., & Rayns, F. (2021a). Historical changes in the mineral content of fruit and vegetables in the UK from 1940 to 2019: a concern for human nutrition and agriculture. *International Journal of Food Sciences and Nutrition, 73*(3), 315–326. https://doi.org/10.1080/09637486.2021.1981831
34. Medline Plus. (n.d.). *Carbohydrates.* Retrieved November 04 2024, from https://medlineplus.gov/carbohydrates.html#:~:text=Glucose%2C%20or%20blood%20sugar%2C%20is,and%20muscles%20for%20later%20use.
35. Mohanta, B., Sen, D. J., Mahanti, B., & Nayak, A. K. (2022). Antioxidant potential of herbal polysaccharides: An overview on recent researches. *Sensors International, 3*, 100158. https://doi.org/10.1016/j.sintl.2022.100158
36. Pan, S., Nie, Q., Tai, H., Song, X., Tong, Y., Zhang, L., Wu, X., Lin, Z., Zhang, Y., Ye, D., Zhang, Y., Wang, X., Zhu, P., Chu, Z., Yu, Z., & Liang, C. (2022). Tea and tea drinking: China's outstanding contributions to the mankind. *Chinese Medicine, 17*(1). https://doi.org/10.1186/s13020-022-00571-1
37. Pang, G., Li, F., Yan, Y., Zhang, Y., Kong, L., Zhu, P., Wang, K., Zhang, F., Liu, B., & Lu, C. (2019). Herbal medicine in the treatment of patients with type 2 diabetes mellitus. *Chinese Medical Journal/Chinese Medical Journal, 132*(1), 78–85. https://doi.org/10.1097/cm9.0000000000000006
38. Prisdiany, Y., Puspitasari, I. M., Putriana, N. A., & Syamsunarno, M. R. a. A. (2021). Potensi tanaman herbal antidiabetes untuk minuman obat: Sebuah literatur review. *Jurnal Farmasi Klinik Indonesia, 10*(2), 144. https://doi.org/10.15416/ijcp.2021.10.2.144
39. Patel, A.R., Piyush V., Chandni, V. Mori, & Parmar V.K. (2023). Effect of fertility levels with micronutrient fortification and bio enhancer on growth and yield of onion (Allium cepa L.). biological forum. *An International Journal, 15*(12), 34-44. https://www.researchgate.net/publication/379669157_Effect_of_Fertility_Levels_with_Micronutrient_Fortification_and_Bio_Enhancer_on_Growth_and_Yield_of_Onion_Allium_cepa_L
40. Rahman, M. M., Islam, M. R., Shohag, S., Hossain, M. E., Rahaman, M. S., Islam, F., Ahmed, M., Mitra, S., Khandaker, M. U., Idris, A. M., Chidambaram, K., Emran, T. B., & Cavalu, S. (2022). The multifunctional role of herbal products in the management of diabetes and obesity: A comprehensive review. *Molecules Annual, 27*(5), 1713. https://doi.org/10.3390/molecules27051713

41. Red Blossom Tea Company. How to brew tea. (n.d.). https://redblossomtea.com/pages/brewing-guide
42. Romano, D. E. U., Putot, F. L. E., Pael, S. L. M., Cordova, J. N. A., Bucog, P. S. B., & Destajo, K. L. M. (2024). Euphorbia hirta (asthma plant) gummies as alternative supplements: An experimental research. *Cognizance Journal of Multidisciplinary Studies, 4*(5), 67–74. https://doi.org/10.47760/cognizance.2024.v04i05.006
43. Sarkar, R. D., & Roy, G. S. (2023). Micro-qualification by estimation of leaf constants of the ethnobotanical plant Costus igneus N.E. Br by optical microscopical analysis. *World Journal of Biology Pharmacy and Health Sciences, 13*(3), 120–170. https://doi.org/10.30574/wjbphs.2023.13.3.0130
44. Sheng, X., Zan, J., Jiang, Y., Shen, S., Li, L., & Yuan, H. (2023). Data fusion strategy for rapid prediction of moisture content during drying of black tea based on micro-NIR spectroscopy and machine vision. *Optik, 276*, 170645. https://doi.org/10.1016/j.ijleo.2023.170645
45. Sivasubramaniam, & Sinnathurai. (2025, April 2). *Tea. Encyclopedia Britannica.* https://www.britannica.com/topic/tea-beverage
46. Song, F., Wu, J., Liu, B., Jiang, J., Li, Z., Song, C., Li, J., & Jin, G. (2021b). Intelligent green tea fixation with sensor fusion technology. *Journal of Food Engineering, 317*, 110846. https://doi.org/10.1016/j.jfoodeng.2021.110846
47. Sen, D. B., Balaraman, R., Sen, A. K., Zanwar, A., Greeshma, K. P., & Maheshwari, R. (2023). Anti-diabetic activity of herbal remedies. *Journal of Natural Remedies*, 373–381. https://doi.org/10.18311/jnr/2023/32182.
48. Shinde, S., Surwade, S., Sharma, R., & Shinde, S. (2022). Costus ignus: insulin plant and it's preparations as remedial approach for diabetes mellitus. *International Journal of Pharmaceutical Sciences and Research, 13*(4). https://doi.org/10.13040/ijpsr.0975-8232.13(4).1551-58
49. Selvakumarasamy, S., Rengaraju, B., Arumugam, S. A., & Ramalakshmi, K. (2021). Costus pictus–transition from a medicinal plant to functional food: A review. *Future Foods, 4*, 100068. https://doi.org/10.1016/j.fufo.2021.100068
50. Singh, S. (2019). *ERBA TII-antioxidant rich and anti-diabetic herbal tea.* Academic. https://www.academia.edu/71550789/ERBA_TII_antioxidant_rich_and_anti_diabetic_herbal_tea
51. Encyclopedia Britannica (n.d.). *Tea tree Britannica.* Retrieved September 20, 2024, from https://www.britannica.com/plant/tea-tree
52. *Tea making process.* (n.d.). Twinings North America. Retrieved April 05, 2024, from https://twinings.ca/pages/tea-making-process
53. Tolentino, R., Van, C., Tomas, J., Travezonda, B., & Magnaye. (2019). Herbal medicine utilization among Batangueños. *Asian Pacific Journal of Education, Arts and Sciences, 6*(1), 9–22. https://research.lpubatangas.edu.ph/wp-content/uploads/2019/07/APJEAS-2019.6.1.02.pdf
54. Vijay, K. V., Kakasaheb, L.G., & Vaibhav, J. (2023). Medical research report of the insulin plant and other herbal plants. *International Research Journal of Modernization in Engineering Technology and Science, 7874*, 2582–5208. Retrieved April 10, 2025, from https://www.irjmets.com/uploadedfiles/paper//issue_5_may_2023/40293/final/fin_irjmets1685379796.pdf
55. Wagh, H. M., Sonwane, N., Suryawanshi, A., & Mairal, P. B. (2022). The review on insulin plant: a plant of ayurvedic used (Costus igneus plant). *International Journal for Multidisciplinary Research, 4*(6). https://doi.org/10.36948/ijfmr.2022.v04i06.1019
56. Wangiyana, I. G. A. S., Sawaludin, Nizar, W. Y., & Wangiyana, W. (2019). Tannin concentrations of gyrinops tea with different leaf processing methods and addition of herbal medicine ingredients. *Proceeding of the 2nd international conference on bioscience, biotechnology, and biometrics 2019*. https://doi.org/10.1063/1.5141326
57. Wang, X., Liu, Y., Gong, N., Wang, J., Han, Z., & Wu, Z. (2024). Design of a tea strip-lining degree discrimination system based on appearance characteristics and near-infrared spectroscopy. *Journal of Food Composition and Analysis*, 106779. https://doi.org/10.1016/j.jfca.2024.106779
58. World Health Organization. (2020). *Guideline: Sugars intake for adults and children.* https://www.who.int/publications/i/item/9789241549028
59. World Health Organization. (2024). *Diabetes.* https://www.who.int/news-room/fact-sheets/detail/diabetes
60. Xie, Y., Gou, L., Peng, M., Zheng, J., & Chen, L. (2021). Effects of soluble fiber supplementation on glycemic control in adults with type 2 diabetes mellitus: A systematic review and meta-analysis of randomized controlled trials. *Clinical nutrition (Edinburgh, Scotland), 40*(4), 1800–1810. https://doi.org/10.1016/j.clnu.2020.10.032
61. Xie, J., Wang, L., Deng, Y., Yuan, H., Zhu, J., Jiang, Y., & Yang, Y. (2023). Characterization of the key odorants in floral aroma green tea based on GC-E-Nose, GC-IMS, GC-MS and aroma recombination and investigation of the dynamic changes and aroma formation during processing. *Food Chemistry*, 427, 136641. https://doi.org/10.1016/j.foodchem.2023.136641
62. Xu, X., Zhao, C., Li, B., Tang, G., Shang, A., Gan, R., Feng, Y., & Li, H. (2021). Effects and mechanisms of tea on obesity. *Critical Reviews in Food Science and Nutrition, 63*(19), 3716–3733. https://doi.org/10.1080/10408398.2021.1992748
63. Ye, J., Ye, Y., Yin, J., Jin, J., Liang, Y., Liu, R., Tang, P., & Xu, Y. (2022). Bitterness and astringency of tea leaves and products: Formation mechanism and reducing strategies. *Trends in Food Science & Technology, 123*, 130–143. https://doi.org/10.1016/j.tifs.2022.02.031
64. Zhang, B., Yue, R., Huang, X., Wang, Y., Jiang, Y., & Chin, J. (2019). Effect of herbal tea on glycemic control in patients with type 2 diabetes. *Medicine, 98*(50), e18346. https://doi.org/10.1097/md.0000000000018346

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved

65. Zhang, J., Wang, J., Li, M., Guo, S., & Lv, Y. (2022). Effects of heat treatment on protein molecular structure and in vitro digestion in whole soybeans with different moisture content. *Food Research International*, *155*, 111115. https://doi.org/10.1016/j.foodres.2022.111115

66. Zhang, Q., Zhao, Y., Zhang, M., Zhang, Y., Ji, H., & Shen, L. (2021). Recent advances in research on vine tea, a potential and functional herbal tea with dihydromyricetin and myricetin as major bioactive compounds. *Journal of Pharmaceutical Analysis*, *11*(5), 555–563. https://doi.org/10.1016/j.jpha.2020.10.002

67. Zhang, X., Zhang, L., Zhang, B., Liu, K., Sun, J., Li, Q., & Zhao, L. (2022). Herbal tea, a novel adjuvant therapy for treating type 2 diabetes mellitus: A review. *Frontiers in Pharmacology*, *13*. https://doi.org/10.3389/fphar.2022.982387

©2025, Cognizance Journal, ZAIN Publications, Fridhemsgatan 62, 112 46, Stockholm, Sweden, All Rights Reserved